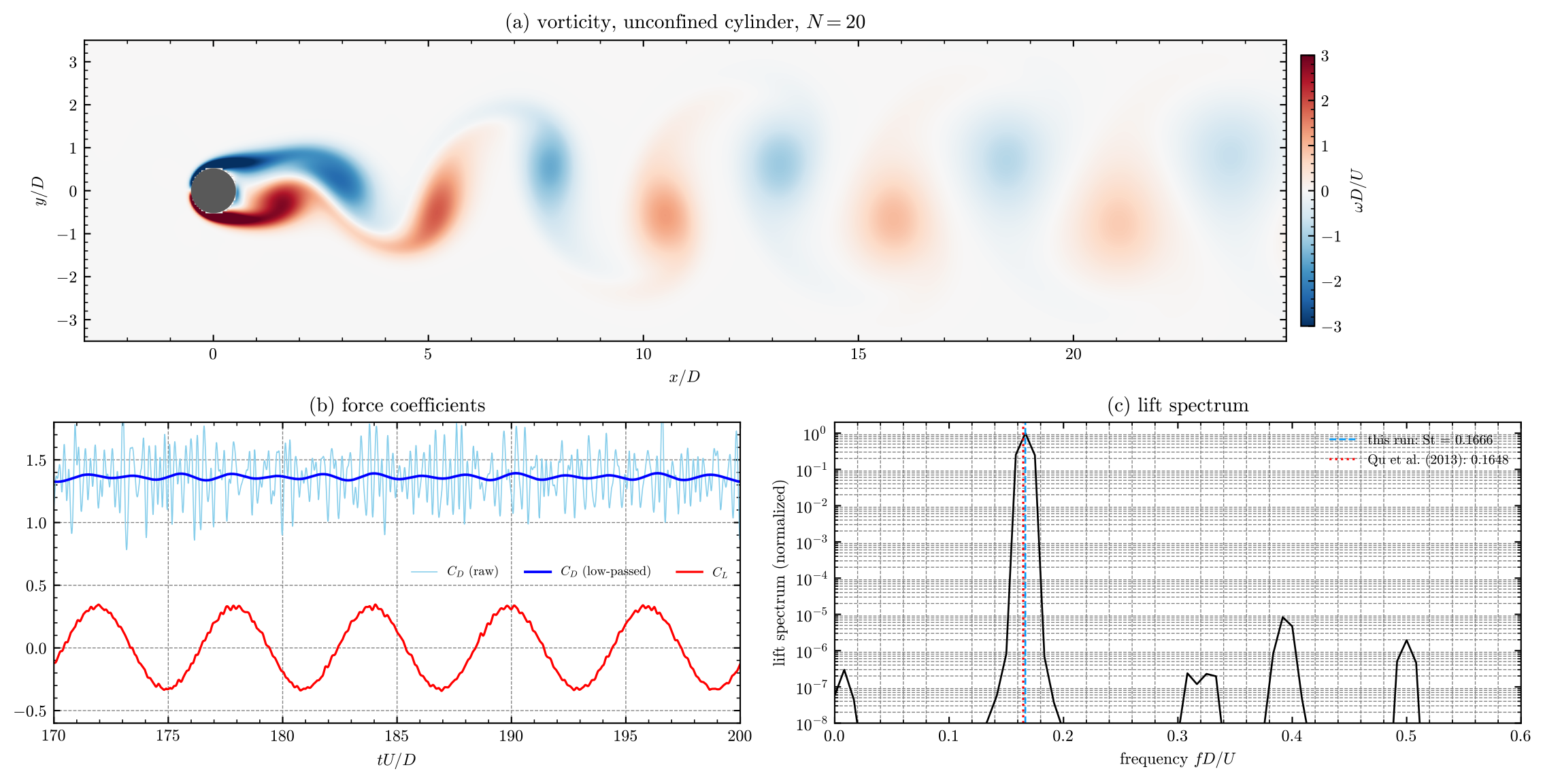

(a) vorticity, unconfined cylinder, $N = 20$
y/D
x/D
$\omega D/U$
(b) force coefficients
C_D (raw)
C_D (low-passed)
C_L
tU/D
(c) lift spectrum
this run: St = 0.1666
Qu et al. (2013): 0.1648
lift spectrum (normalized)
frequency fD/U